



Shazam now faster!

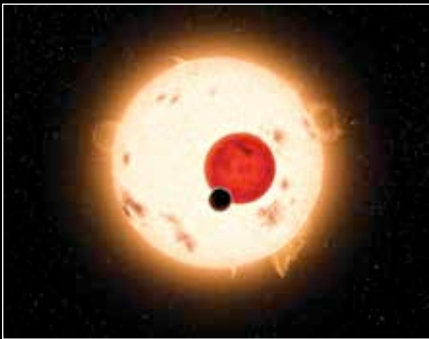
After the latest update, Shazam's song recognition now works even faster. Know more: <http://dgit.in/FastShazam>



Google's new badge!

Google employees are working on a communicator badge like Star Trek to communicate with just a touch. Read here <http://dgit.in/GCbadge>

Feature



Hello Tatooine, two very good mornings to you

it would still need a weeks supply to blow up Alderaan. The Death star didn't need a week. It needed a few seconds tops. Or maybe they used antimatter.

Thanks to Einstein we know that if we manage to convert matter to energy we can obtain a large amount of it. Antimatter, as its name suggests mirrors ordinary matter in regard to some properties. If antimatter collides with regular matter it results in mutual annihilation, releasing energy in the process. Some number crunching gives us an estimate for the mass of antimatter needed to blow Alderaan – a trillion tonnes. In the grand scheme of things that is not really very much and it is very likely that that much antimatter can be harvested and safely stored. Current day earthlings haven't figured out a way to keep stored antimatter away from normal matter, but then we haven't made a lightsaber either. So there you go. The inventors of the Death Star probably figured a way to stream antimatter. The antimatter met Alderaan's ordinary matter and boom!

The million dollar question is does any Earthbound technology come close to the Death Star's power? Nope. Not even kind of close. Phew!

Lightsaber – Enough said

A lightsaber is easily the most famous weapon there is : real or not. We're pretty sure there are people in Honolulu who haven't heard of an AK-47 but mention a lightsaber and they'll high five you for sure. A lightsaber is elegant, powerful, brilliant and awesome. Those are four words you never hear being associated with a weapon, do you? But, the lightsaber is all of that and more. What's not to like about a powerful

beam of contained energy that can literally cut through anything and one that you can wield like a freaking sword! If you didn't know better you would call it a laser stick (you'd of course be decapitated for that travesty). Still the lightsaber does look like a fancy laser, yes, but it's highly unlikely that the beam of energy is a laser. A laser is basically a controlled beam of electromagnetic radiation. For a "laser" lightsaber you would have to have a contraption that emits a beam of high intensity light and somehow at the end of a metre stop it. You see, light doesn't do that. It just keeps going, traveling at a massive speed unless it is reflected, absorbed or... wait a minute, *pulled back*. What has the gravitational pull to pull light? You guessed it right, its a black hole. Yes, those mysterious powerful beasts would somehow have to exist in in a lightsaber hilt to have a lasersaber. Of course this causes some practical issues with usability. A black hole will swallow all matter inside a radius, not to mention completely swallow the emitted laser beam. Also to fit inside the hilt, a black hole will have to have a small volume which means it would have to be heavy, really really heavy. If CERN found out a way of producing micro black holes that could be tuned or controlled this bizarre contraption might actually work, but we have another problem. Light is made up of photons which when interacting with other photons (read : second light saber) would really not behave like a solid (as in the films), instead just passing through unscathed. Should this debunk the whole laser based lightsaber theory? Maybe not.

Very recently some physicists from MIT and Harvard succeeded in defying what



Dear CERN how are those portable black holes coming along? We need our lightsabers please.

was known about photon interactions and clumped photons to form some sort of a pseudo molecule boosting the chance of someday using light to actually form a solid like structure. Scientists have succeeded in slowing down light considerably and recently a beam of light was actually trapped in a crystal for almost a minute. Maybe, just maybe, one day scientists will succeed in combining these techniques and come up with a laser based lightsaber. Till then however we would have to let a lasersaber be, no matter how cool a portable black hole sounds.

Can Physics offer some other explanation. Yes. It almost always can. Perhaps the energy beam of the lightsaber is made up of plasma, often referred to the as the fourth state of matter. Plasma is the state created when we superheat a gas to such an extent the atomic electron bonds break releasing charged ions that are very conducting. Plasma torches that can cut through common day materials have been made. But those are unwieldy, have small beams

and require large amounts of energy. So, how can a plasma based lightsaber work? Here we go. The hilt of the saber houses all the advanced technology needed to guide, contain and power the saber. A powerful battery produces the plasma state which is then focused and passed via a crystal based optical setup to generate the saber's unique glow and properties. The beam is



We just told you how to make one. What're you waiting for?